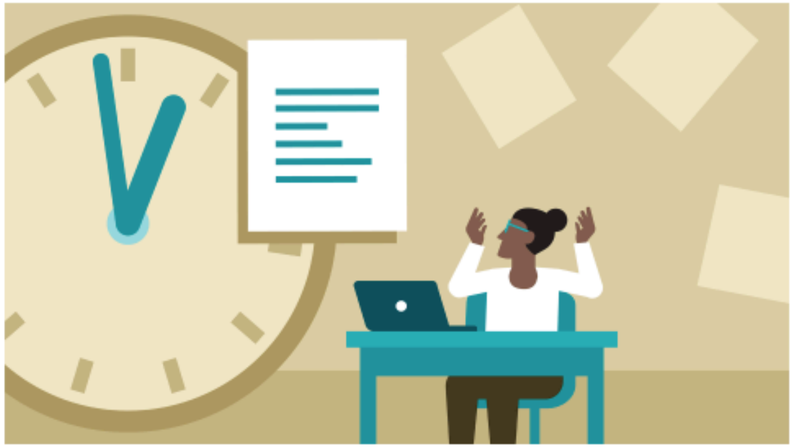
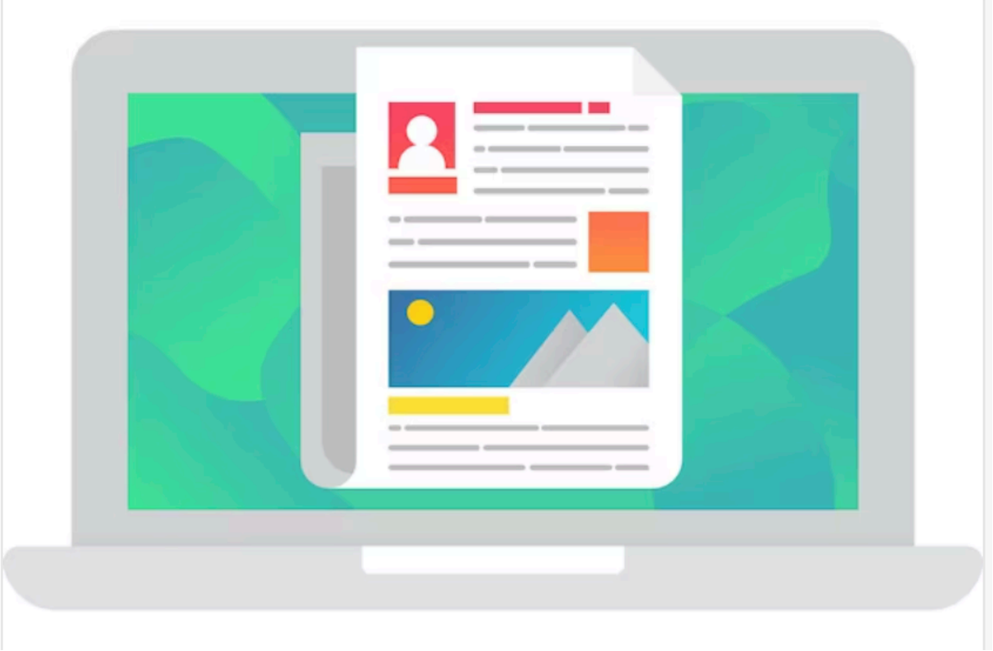
Assignment 2 Specification

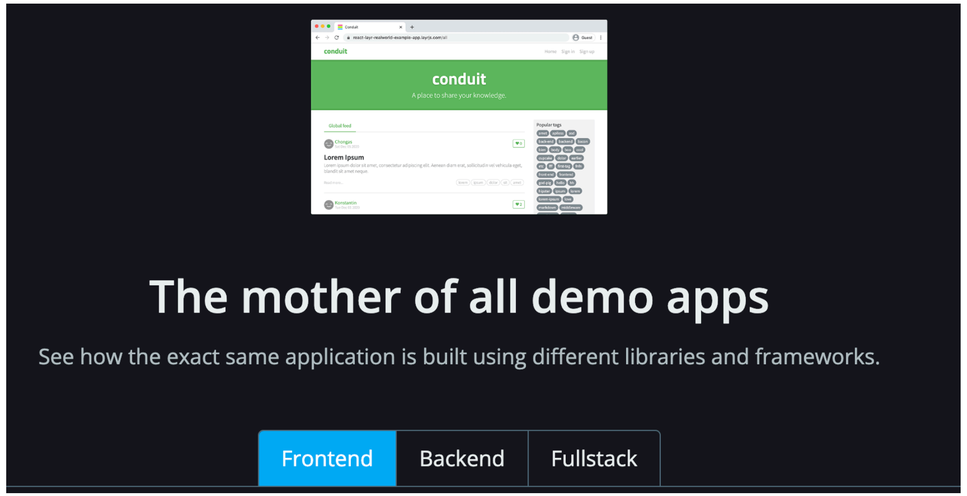


Concept, detail and grading spectrum for Assignment 2

Assignment Motivation & Philosophy

- Assignments present multiple opportunities:
  - Crystallise knowledge on module
  - Opportunity for creativity & innovation
  - Potentially important portfolio contribution
- It is not over when submitted, presented and graded
  - Ongoing maintenance & enhancements
  - Periodically rebuild using new technology stack





https://github.com/gothinkster/realworld
https://codebase.show/projects/realworld

Agenda

Full Stack 1 Technology Platform

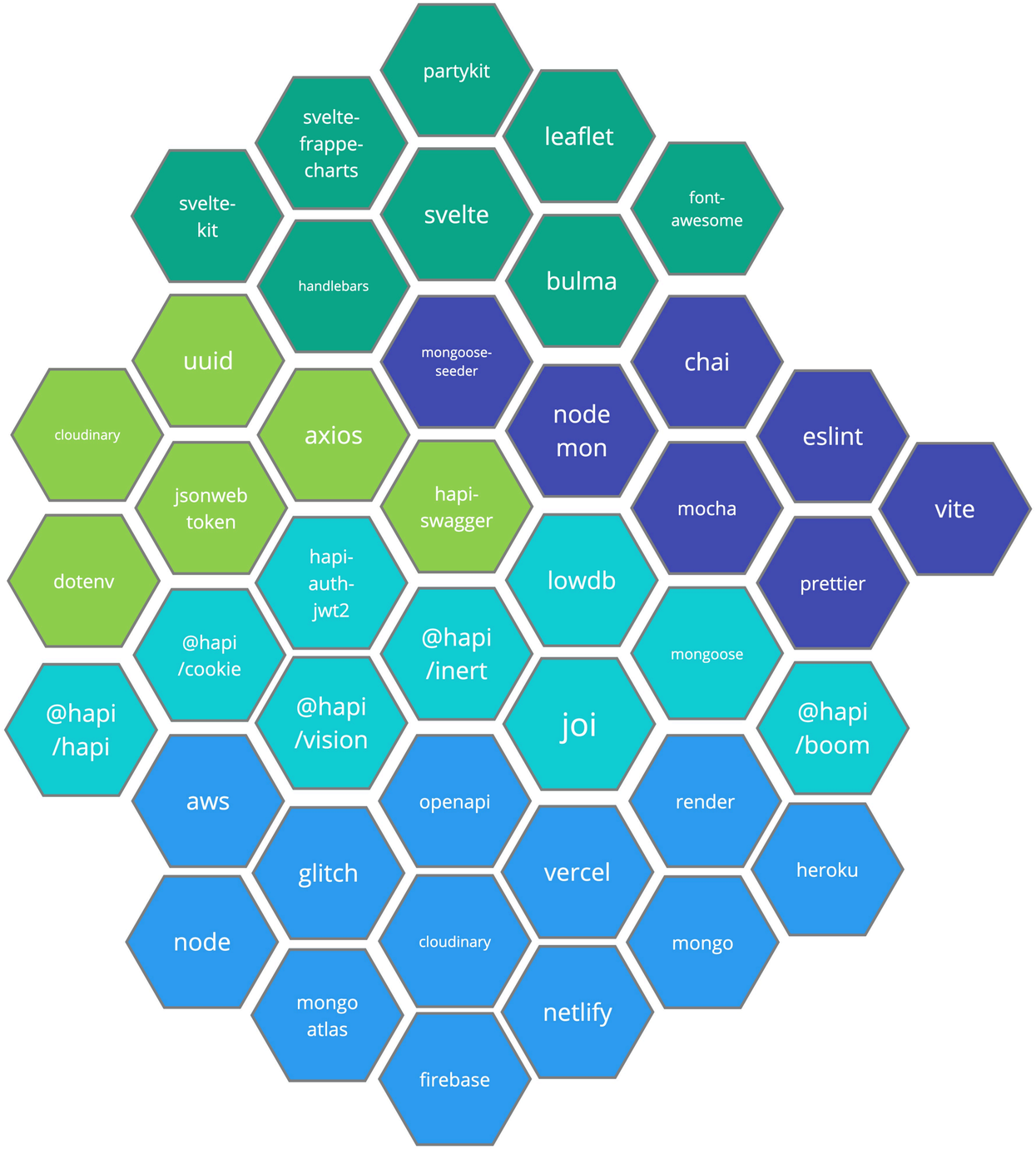
Assignment Briefing

Assignment Versions (choice)

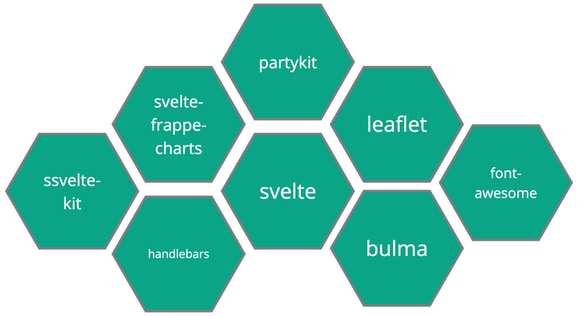
Grading Rubric(s)

Full Stack Technology Platform

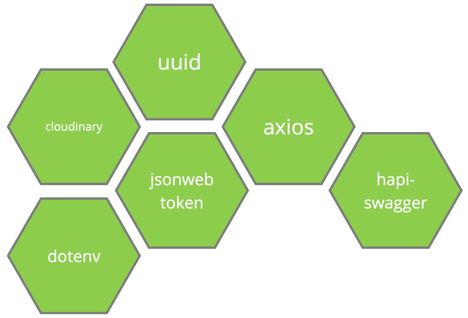
Full Stack Components & Services



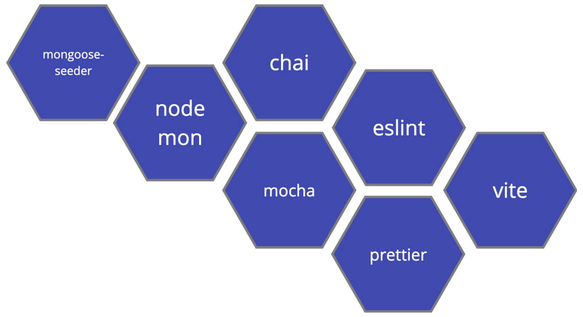
Front End



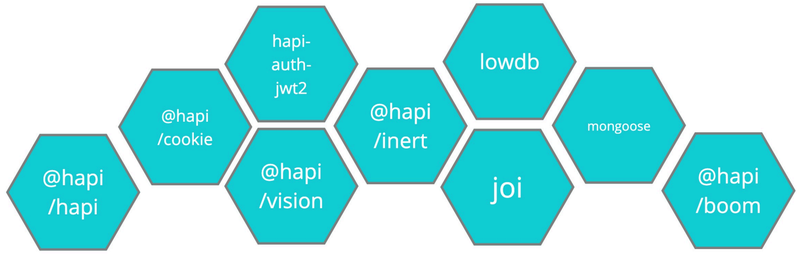
Components



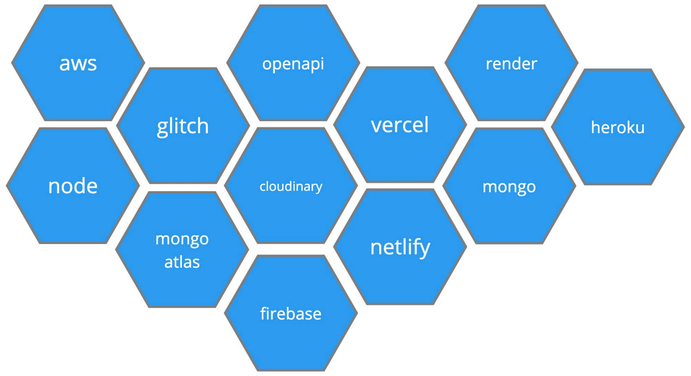
Tools



Back End



Infrastructure



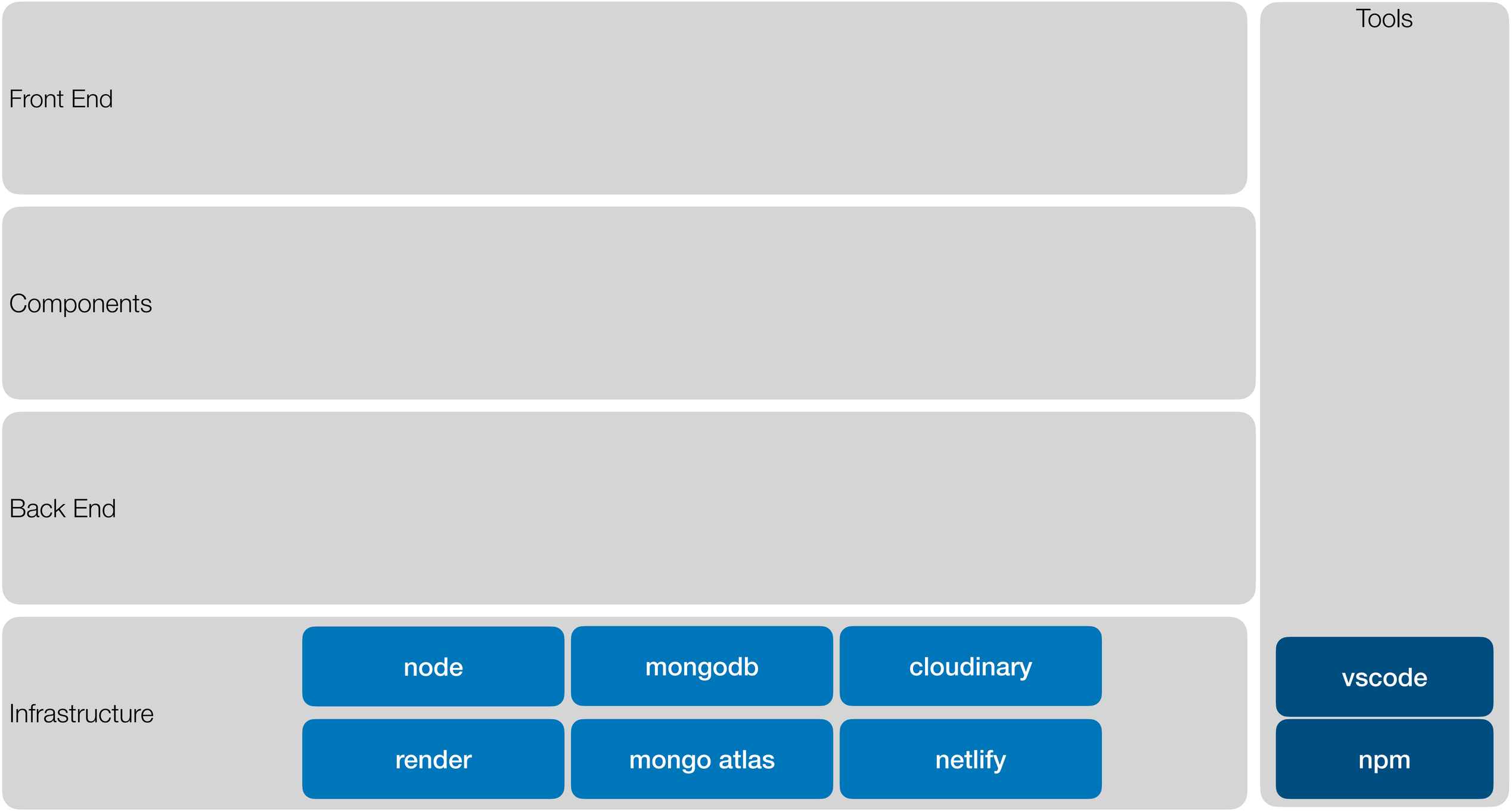
Front End

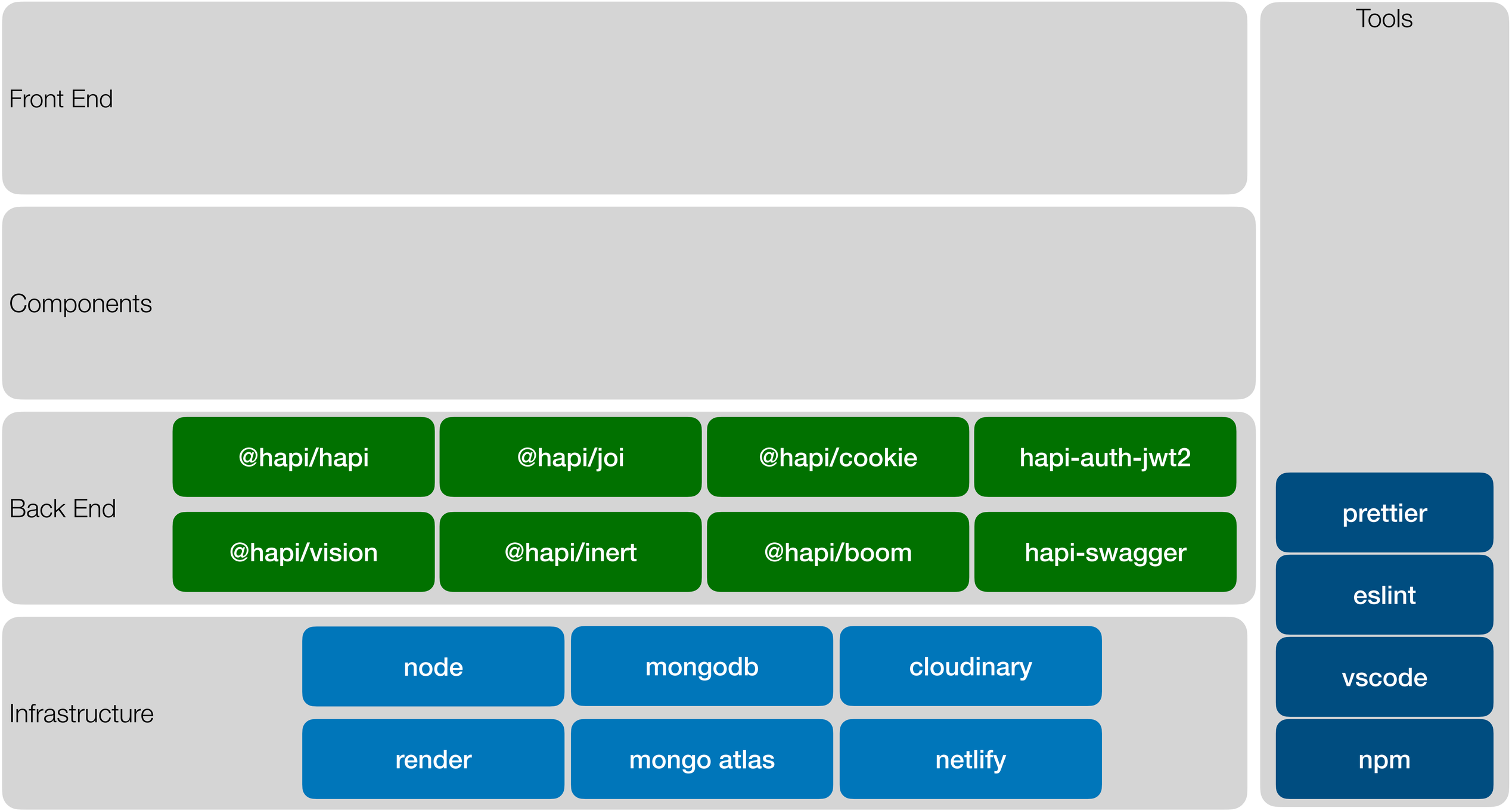
Components

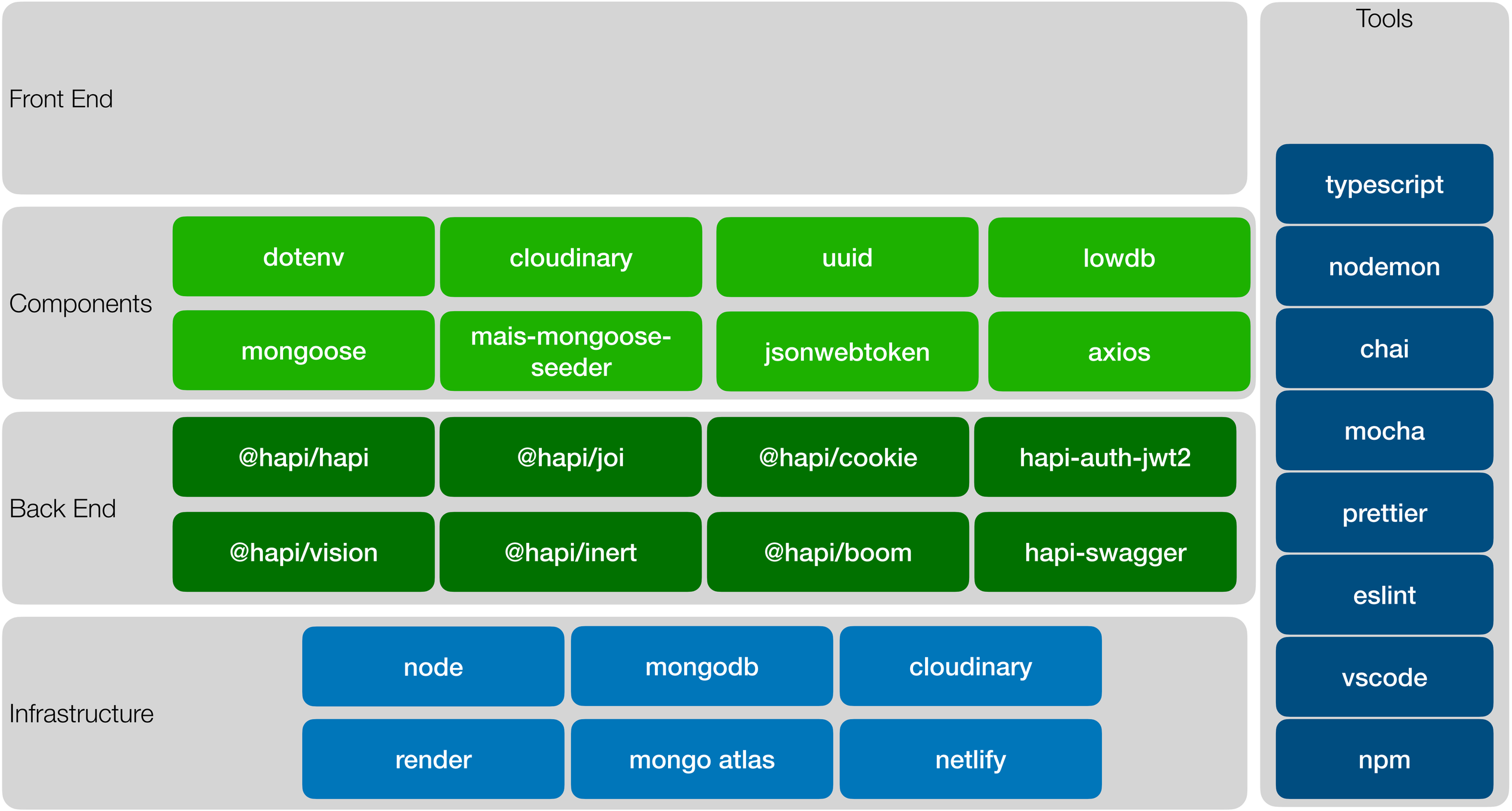
Back End

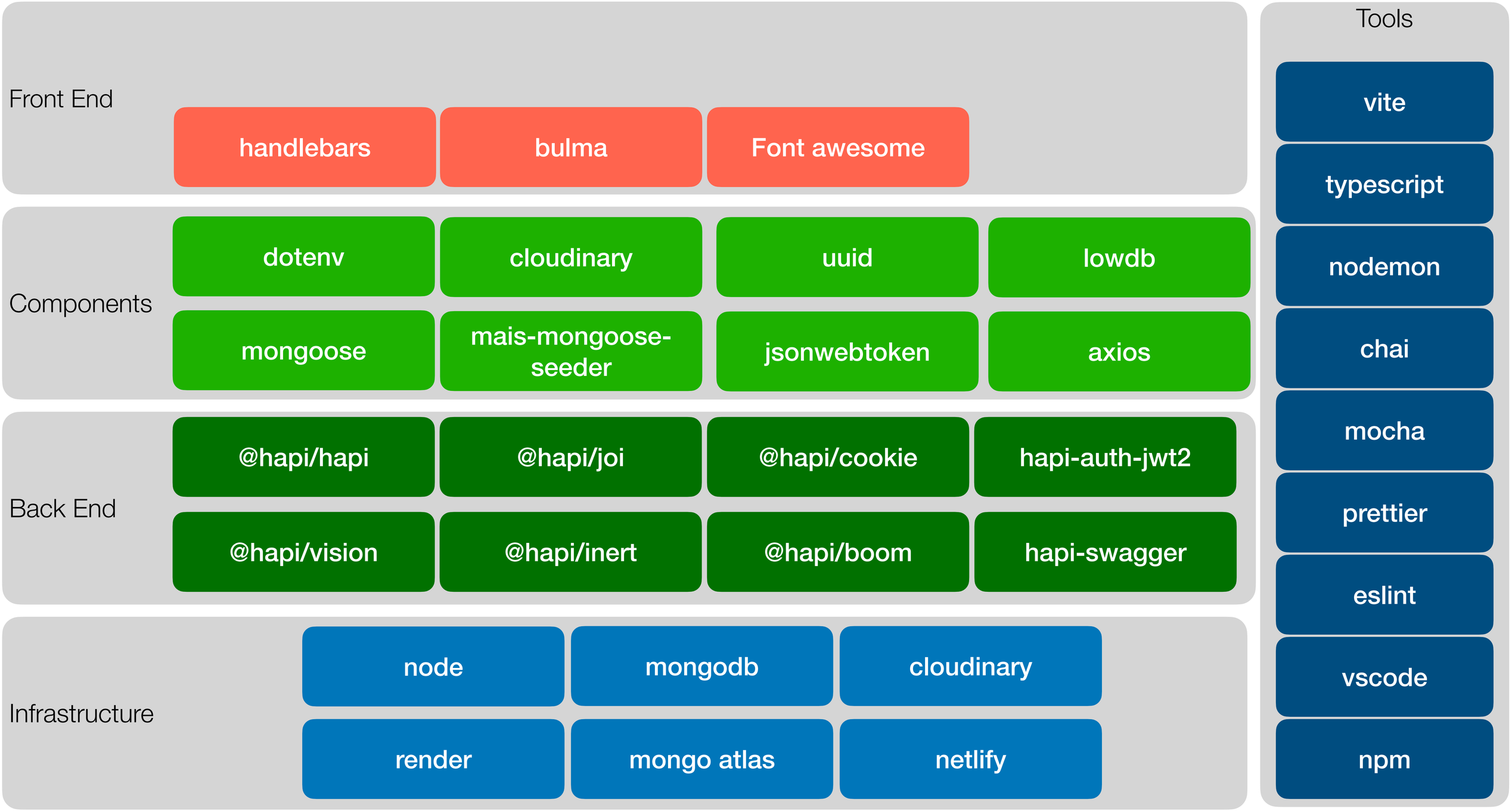
Infrastructure

Tools


Front End
Components
Back End
Infrastructure
node
mongodb
cloudinary
render
mongo atlas
netlify
Tools
vscode
npm


Front End
Components
Back End
@hapi/hapi
@hapi/joi
@hapi/cookie
hapi-auth-jwt2
@hapi/vision
@hapi/inert
@hapi/boom
hapi-swagger
Infrastructure
node
mongodb
cloudinary
render
mongo atlas
netlify
Tools
prettier
eslint
vscode
npm


Front End
Components
dotenv
cloudinary
uuid
lowdb
mongoose
mais-mongoose-seeder
jsonwebtoken
axios
Back End
@hapi/hapi
@hapi/joi
@hapi/cookie
hapi-auth-jwt2
@hapi/vision
@hapi/inert
@hapi/boom
hapi-swagger
Infrastructure
node
mongodb
cloudinary
render
mongo atlas
netlify
Tools
typescript
nodemon
chai
mocha
prettier
eslint
vscode
npm


Front End
handlebars
bulma
Font awesome
Components
dotenv
cloudinary
uuid
lowdb
mongoose
mais-mongoose-seeder
jsonwebtoken
axios
Back End
@hapi/hapi
@hapi/joi
@hapi/cookie
hapi-auth-jwt2
@hapi/vision
@hapi/inert
@hapi/boom
hapi-swagger
Infrastructure
node
mongodb
cloudinary
render
mongo atlas
netlify
Tools
vite
typescript
nodemon
chai
mocha
prettier
eslint
vscode
npm

Front End	Svelte	sveltekit	leaflet	frappe-charts
	handlebars	bulma	Font awesome	iconify
Components	dotenv	cloudinary	uuid	lowdb
	mongoose	mais-mongoose-seeder	jsonwebtoken	axios
Back End	@hapi/hapi	@hapi/joi	@hapi/cookie	hapi-auth-jwt2
	@hapi/vision	@hapi/inert	@hapi/boom	hapi-swagger
Infrastructure	node	mongodb	cloudinary	
	render	mongo atlas	netlify	

Tools

- vite
- typescript
- nodemon
- chai
- mocha
- prettier
- eslint
- vscode
- npm

Assignment 2 Briefing

Point of Interest

A point of interest, or POI, is a specific point location that someone may find useful or interesting.

"A "point of interest" (POI) is a location for which information is available. A POI can be as simple as a set of coordinates, a name, and a unique identifier, or more complex such as a three-dimensional model of a building with names in multiple languages, information about opening and closing hours, and a civic address. POI data has many applications, including augmented reality browsers, location-based social networking games, geocaching, mapping and navigation systems."

http://www.opengeospatial.org/projects/groups/poiswg

https://en.wikipedia.org/wiki/Point_of_interest

https://en.wikipedia.org/wiki/Point_of_interest

PlaceMark Examples

- Landscape feature
- National monument
- Walking Trail
- Bridge
- Tree
- Venue
- Ringfort
- Dolmen
- River
- Bog
- Island
- Forest
- Beach
- Museum
- Climb
- Cycleway
- Birdwatching Locations
- Diving

Product Feature Sets

POI Sharing / Community Features

Community

- You are free to innovate an the type, range and variety of features

Richer Media Experience + User Management

Enhanced

- These categories are to stimulate your brainstorming

Essential Data Structure & Support Service

Core

- Some ideas on the next slides…

Information Model

- Placemark
  - Name
  - Category
  - Description
  - Analytics
- User
  - Basic user details

Features

- Sign up / log in to service
- Create, Read, Update, Delete
- Query
- Group into Categories

Core

Information Model

- Placemark
  - Images
  - Location
  - Weather
- User
  - Types

Features

- Placemark images
- Placemark Map Location
- Admin
  - Admin user
  - Admin dashboard

Enhanced

Information Model

- Placemark
  - Creator / Editor
  - Rating
  - Reviews
  - Discussion
- User
  - Interaction information

Features

- Share
- Review & Rate
- Notice Board
- Live Updates
- Analytics

Community

Assignment Versions

PlacemarkCore

Assignment 1

PlacemarkHAPI

OR

PlacemarkSveltekit

PlacemarkCore

You have already completed a version of this

Implement POI model

Support the following User Management Features:

- Sign up / Login in / Delete Account
- Simple User Admin Dashboard

Support the following POI Characteristics:

- Name, Description, Category, Location, Image

Support the following app features:

- Create, Read, Update & Delete POIs
- Organise PlaceMarks into categories

API

- Endpoints + tests
- Swagger Documentation
- JWT Security

Having completed this

PlacemarkCore

Then proceed with one of

PlacemarkHAPI

?

PlacemarkSveltekit

Option A

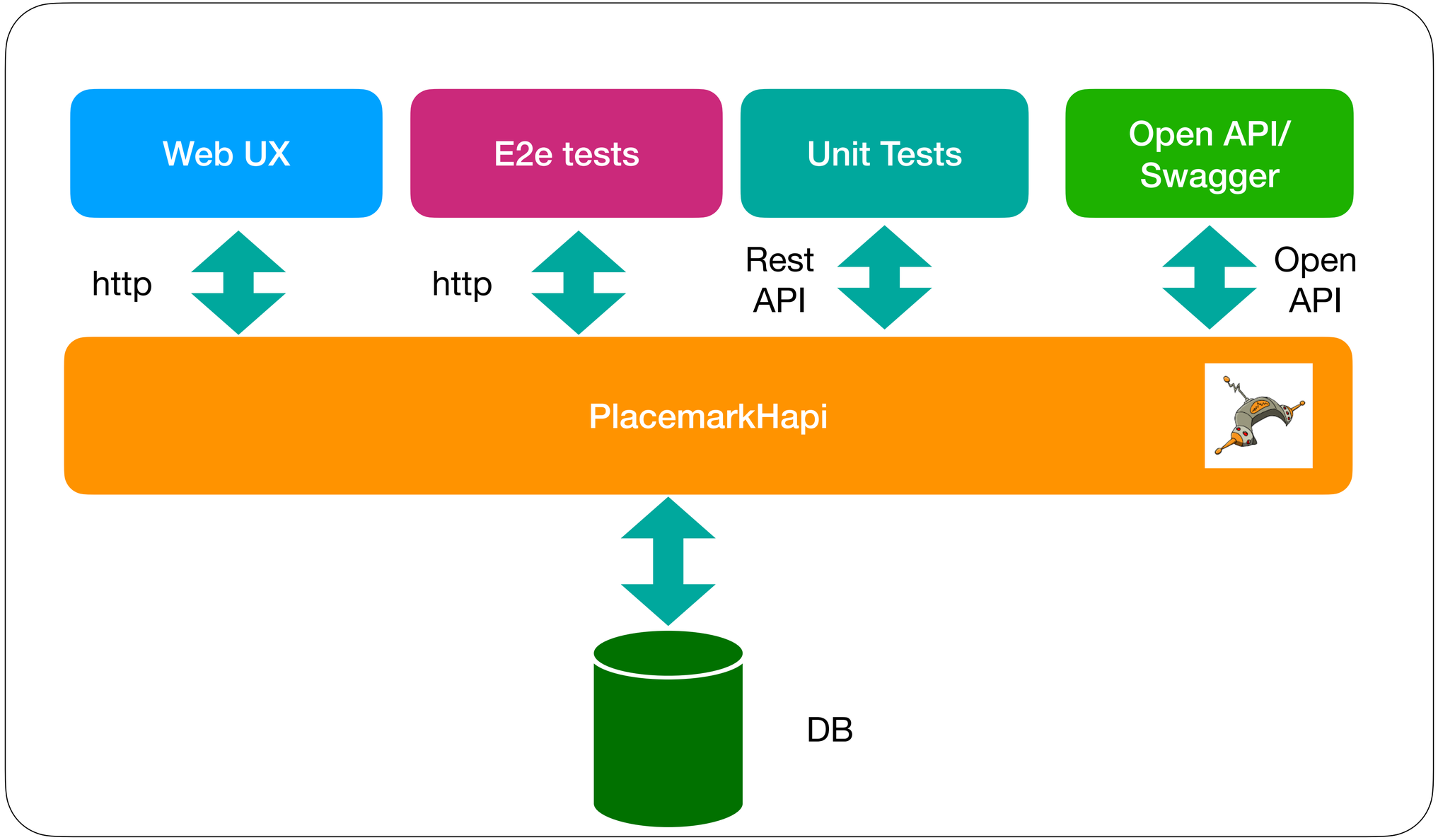
OR

Option B

PlacemarkHAPI

- Choose **PlacemarkHAPI** if you wish to:
  - Deepen your knowledge of Hapi development
  - Implement more sophisticated Models
  - Pursue TDD in more depth
  - Explore End-to-End testing (independently)
  - Continue to evolve the features of your application

PlacemarkHAPI Architecture



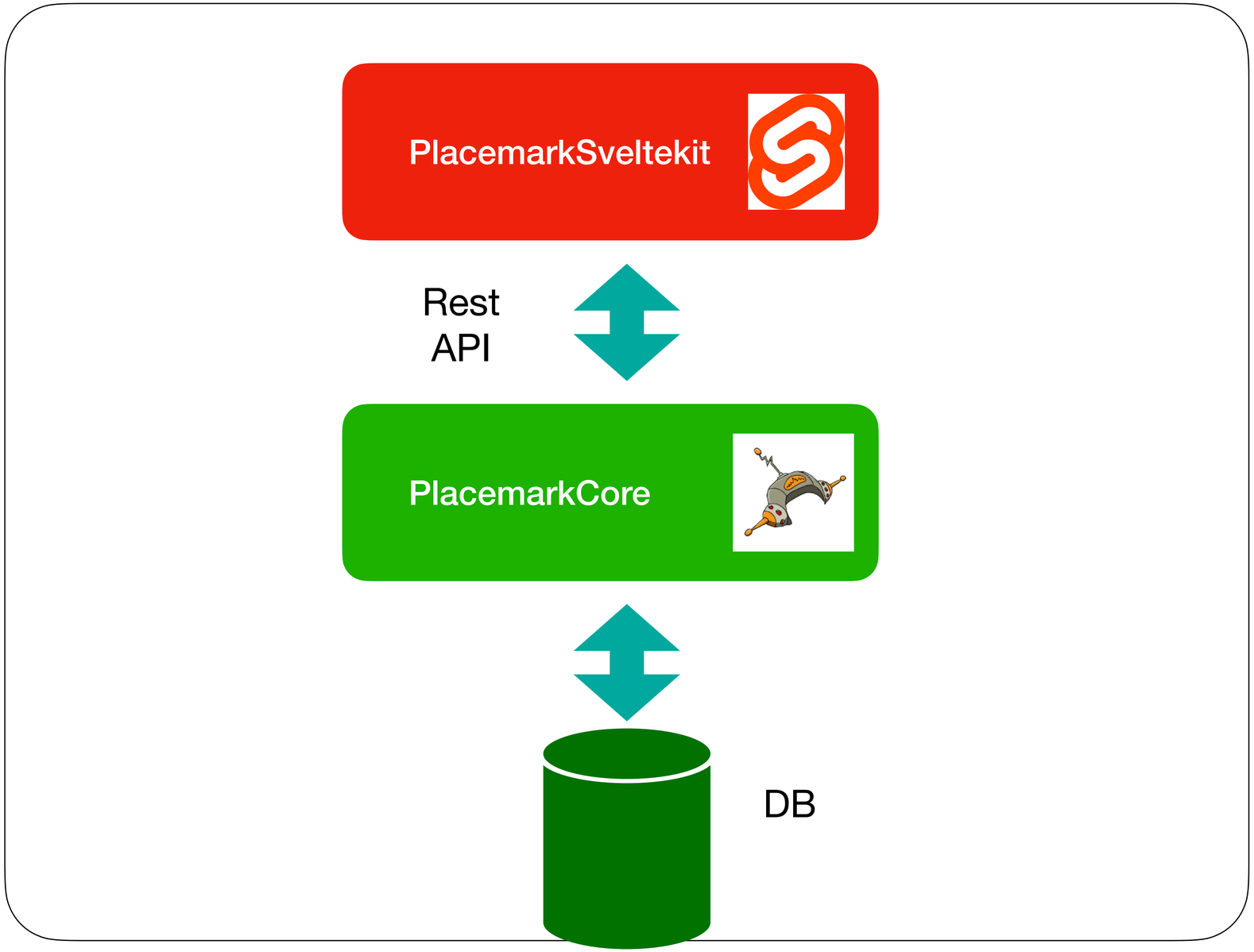
PlacemarkHAPI

	Social 1	Social 2	TDD	Authentication	Devops
Level 1	Private POIs	Reviews	Unit Tests	Sanitisation of input & output	Tagged Releases
Level 2	Public POIs	Share	Unit tests + coverage reports	Hashing & salting passwords	Git Flow
Level 3	Rating	Noticeboard	e2e Playwright - Basic	OAuth (login via Github, Google etc)	Render, AWS
Level 4	Favourites	Discussions	e2e Playwright - Enhanced	3rd Party Auth Service (Auth0, Firebase or equivalent)	Typescript

PlacemarkSveltekit

- Choose **PlacemarkSveltekit** if you wish to:
  - Gain an understanding of modern front end development
  - Develop an engaging user experience
  - Experiment with compelling Mapping features
  - Explore interesting charting components
  - Feature freeze your assignment 1 app (more or less)

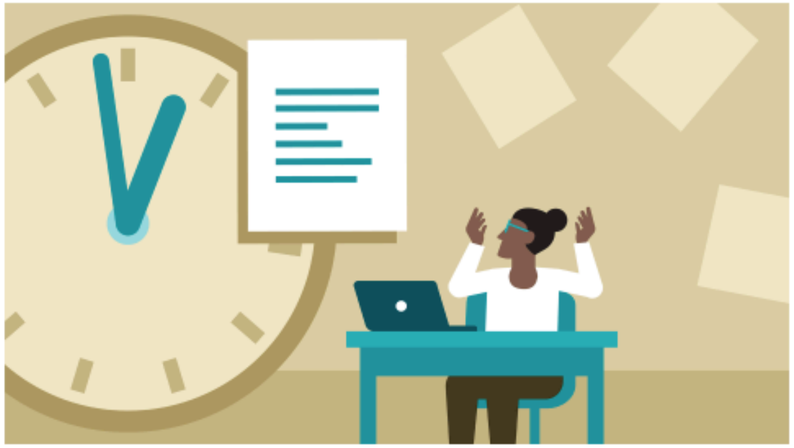
PlacemarkApp Architecture



PlacemarkApp

	Analytics	Maps	Images	Authentication	Architecture
Level 1	Single Chart	Show POI on Map	Single Images per POI	Sanitisation of input & output	Svelte
Level 2	Multiple Chart Types	Map Layers based on Categories	Support Delete Image	Hashing & salting passwords	SvelteKit
Level 3	Apache ECharts (or equivalent)	Multiple Maps on single page	Multiple Images per POI	OAuth (login via Github, Google etc)	SSR
Level 4	Trends and Projections	Weather Conditions at POI / Weather Maps	Image Gallery Feature	3rd Party Auth Service (Auth0, Firebase or equivalent)	Full Stack

Assignment 2 Specification



Concept, detail and grading spectrum for Assignment 2